LIKHA stage 1: technical report (2026-11-06)

Stan Lee V. Natividad
Kooya PH AI Research
leenatividad5@gmail.com

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Status: stage 1 finished and fetched; behaviour evaluation still to do (section 6). INTERNAL model, no release, no served endpoint.

1 What was trained

LIKHA stage 1 is a 354,280,448-parameter decoder-only, Llama-style protein language model trained FROM SCRATCH (own weights; the Llama architecture and the training recipe are public, the base is not anyone else's). 28 layers, hidden 1024, 16 heads (head dim 64), SiLU MLP 2752, RoPE theta 10000, context 1024, tied embeddings, vocabulary 64 with a character-level amino-acid tokenizer and domain tokens (`<dom:agri>`, `<dom:industrial>`, `<dom:general>`; `<dom:health>` reserved and never emitted; no pathogenicity/host/virulence token). Sequence format: `<dom:x><sep><start>RESIDUES<end><|endoftext|>` (tag dropout 10%).

Data (receipt `corpus_sha256 ab5bfe7e...`, 5,564,585 sequences, tagged DOM/SRC/OX): `uniref90_plant` 5,075,542; `ph_crop` 373,403; `industrial_3a_reviewed_proxy` 73,689; `viridiplantae_reviewed` 41,951. Natural-proportion mix, 3 passes. No bacterial/viral class A/B/C data and no gray-zone data (gates parked by Stan until stage 1 ended).

Training: 1x H100 80GB (SECURE), bf16, sdpa attention with document masking, batch 16 x grad-accum 2 x 1024 = 32,768 tokens/step, AdamW lr 3e-4 with 200 warmup steps and a cosine schedule planned for 213,623 steps, seed 1337, save every 1,000 steps.

2 How it ended

Stop reason `DATA_EXHAUSTED` (exit code 5, a designed exit, not a crash) at step 185,373 of the planned 213,623 after 6,074,302,464 tokens: three passes of the corpus (16,610,187 records used) hold fewer tokens than the 7,000,000,000-token budget that set the 213,623-step schedule. This is a PLANNING ERROR on our side (devops): the step budget was derived from a token target the corpus could not supply in 3 passes. Consequences, stated plainly: (a) the learning-rate schedule was cut at 87% (lr 4.2e-05 at the stop, never annealed to its minimum), so the final weights are probably somewhat worse than a completed schedule would give (size of the effect not measured); (b) the run ended about 7 hours earlier than the 32-hour projection and about USD 30 cheaper than the USD 127.5 projected.

Spend: USD 97.63 (training estimate in the receipt), USD 98.06 including the ship window (spend ledger row, 1,474.6 min at USD 3.99/h); the USD 150 ceiling was never approached; one pod existed and was confirmed terminated (GET 404, "pods remaining: 0").

3 Results (measured, from the run's own files)

Step	val_loss	ppl agri	ppl industrial
2,000	2.7075	14.96	18.01
20,000	1.9891	7.24	15.55
60,000	1.8148	6.06	16.26
100,000	1.7218	5.53	13.81
140,000	1.6123	4.99	7.48
170,000	1.5571	4.69	11.18
185,373 (final)	1.5344	4.59	11.09

Stability: no non-finite gradient, 6 clipped steps of 3,799 logged, max grad norm 2.24, peak VRAM 31.6 GB, throughput about 70,000 tokens/s. Final training loss (mean of the last 20 steps) 1.71.

CAVEATS that govern how to read the table: 1. The validation set is tiny and proxy-split: 65,288 tokens in total, of which only 839 tokens are industrial. The "industrial perplexity" that rose, fell, and bounced between 6.8 and 11.6 during training is therefore statistical noise on a few sequences, not a trend (this also means the earlier "industrial drift" alarm that led to the equal-per-source relaunch rested on a very noisy signal). 2. The split is the proxy `OX|first-6-mer` key, not a homology-clustered split; near-homologs can sit on both sides, so val_loss is optimistic. The MMseqs2 0.9/0.8 cluster split does not exist yet. 3. val_loss is over the tagged format, so it is not directly comparable with published protein-LM perplexities. 4. No behaviour evaluation has been run yet (section 6); nothing here says the model generates good proteins.

4 Artifacts (verified)

Fetched via runpodctl; `results.tgz` sha256 `b982b76c99e4789e963a96ab15e8a7b8436d2564efb1cb628893a26efe006e63` equals the value the pod printed. The export MANIFEST sha256 values all match after extraction AND after copying to durable storage:

File	bytes	sha256
model.bf16.safetensors	708,589,824	82fa06618c430e8cb814c8e8063dba73c1fcf2b957fde
model.fp32.safetensors	1,417,150,648	47fc42773566ad300775177d383303afd7eb4f1511b9
config.json	652	f613b4250c0238e5392126ad25c022de370f05542957
tokenizer.json	999	e17efa81861f759a45960e55287154e5f44f6370be724
EXPORT.json	315	ce74b1323a0130dcc4e93772496a4a0354483b79ea68

Location: `D:\PinoAI-data\likha\stage1-step185373\` (export, run receipt and metrics, logs; the 1.9 GB original `results.tgz` is kept under `C:\Users\STAN\.pinoai-runpod\likha-a-350m-full_results\`). The AdamW checkpoints and the corpus were never shipped, by design; only small CKPT.json files came back. Because the optimizer state is not kept, "continue training" (stage 2) restarts the optimizer from the exported weights, which is acceptable but should be stated in the stage-2 plan.

5 Process notes

The first two full launches of this run were stopped by devops (no live progress output; equal-per-source sampling exhausted the small sources early); the third launch (natural-proportion mix) is the run reported here. A separate lesson recorded earlier: fire clients must be detached; they were, and the laptop sleeping during the run did not affect the pod.

6 Behaviour probes (added 2026-11-06, exploratory)

Details and tables: `experiments/bio-behavior-2026-11-05/LIKHA-STAGE1-PROBES.md`. Summary (updated after a corpus membership check against the sha-verified stage-1 corpus, section L4 of that file): probe proteins that are exact TRAIN members are very likely (rice 0.63 nats/residue, actin 0.10) while proteins not in the corpus (p53, haemoglobin, GFP, albumin, insulin) sit at shuffle level (2.8 to 2.9; shuffle 2.95, uniform floor 3.00); on 470 val sequences NLL falls steadily with 8-mer similarity to TRAIN (2.57 plant / 2.64 industrial for novel sequences, perplexity about 13, down to 0.84 / 1.12 for near-duplicates), so the proxy-split val_loss 1.53 is flattered (58% of plant and 43% of industrial val sequences have containment >= 0.3 with TRAIN); the domain tag barely moves likelihood (<= 0.03 nats on rice) but does change sample length and FLORA T2 confidence; 60 samples at T=1.0 look protein-like on surface statistics (Met start, no homopolymers) but nothing was folded; greedy continuation reproduces 84 to 93% of 4/6 rice proteins (all six are exact TRAIN members: memorisation, partial since two of six were not recalled) and 5 to 11% of the three non-plant ones. These probes do not show the model generates good or novel proteins.

7 Generation quality corrected (added 2026-11-06)

The "~3% of samples fold confidently" figure above came from sampling at the harshest setting (T=1.0, top_k=0). A temperature x top-p grid (docs/LIKHA-SURGERY-LOOP-2026-11-06.md, iteration 7) shows that at T=0.75, top_p 0.9 the confident-fold rate (ESMFold pLDDT >= 70) is ~37.5% (mean pLDDT 63), a ~4.5x gain from sampling alone. The recommended generation config and tool are in docs/LIKHA-GENERATION-CONFIG.md and scripts/likha/likha_generate.py. Foldability is also linearly decodable from the model's activations (probe AUROC 0.98), giving an ESMFold-free foldability scorer. So LIKHA's usable-output rate is much higher than the stage-1 harsh-sampling numbers implied; describe outputs as computationally predicted foldable candidates (pLDDT is a proxy, not validation).

8 Next

Behaviour evaluation (planned, exploratory): per-residue likelihood of known plant and non-plant proteins vs shuffled controls; unconditional samples per domain tag (composition, length, repeats, FLORA T1/T2 class distribution); perplexity on a cluster-disjoint split once clusters exist; ESMFold pLDDT of samples (pod or Kaggle). Stage-2 planning inputs: a larger, cluster-disjoint held-out set including enough industrial sequences (the current 839 tokens cannot support any conclusion), a schedule sized to the corpus (or more data) so the lr anneals, and, after the gates Stan parked, the class A/B/C data. Gray-zone pathogen/AMR sequences stay out of generative training (hazard line).